Agency Credit Roll Rate Transition Model

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15th April, 2015

Mortgage Valuation System - Components

- **Voluntary Prepayment Model: projects voluntary speeds (VPR)**
- **Delinquency Roll Rate Model: transitions between states**
- **Term Structure Model: generates simulated yield curve paths**
- **MOATS Model: calculates mortgage rates as a function of yield curve and vols**
- **Home Price Simulation Model: projects home price changes as a function of past home prices and mortgage rates – needed for credit-adjusted OAS analysis**

Integrated Prepayment & Default Models

Prepayment Rate = (Turnover + Refis + Curtailments) + Defaults

= **Voluntary Prepayments + Defaults**

Default Component – function of

- **Loan Features:** Loan Type, WALA, Payment Shock, Downpayment, CLTV, Purpose, Lien Status, Documentation, SATO
- **Borrower Characteristics:** Credit Score (FICO), DTI, Occupancy Status
- **Macro-Economic Variables**: Home Price Appreciation, Home Price Momentum, Unemployment Rates, Interest Rates.

Interaction between **Voluntary** and **Involuntary Prepay** Components:

- **Refi Burnout** – better borrowers leave the pool
- **Credit Burnout** – worse borrowers leave the pool

Framework of the Roll Rate Model

We define the following possible delinquency states for a loan:

- Clean Current: current for past 24 months (CC)
- Dirty Current : current today but missed at least one payment in the past 24 months (CD)
- Early Delinquency: 30-days delinquent (D3)
- Seriously Delinquent: including 60 or more days delinquent (SD)
- Voluntary prepayment (PP)
- Default (DF)

Note that PP and DF are absorbing (or terminal) states resulting in loans exiting the pool. Let $S(t)$ be the delinquency status distribution at time t, and $T(t)$ be the transition matrix

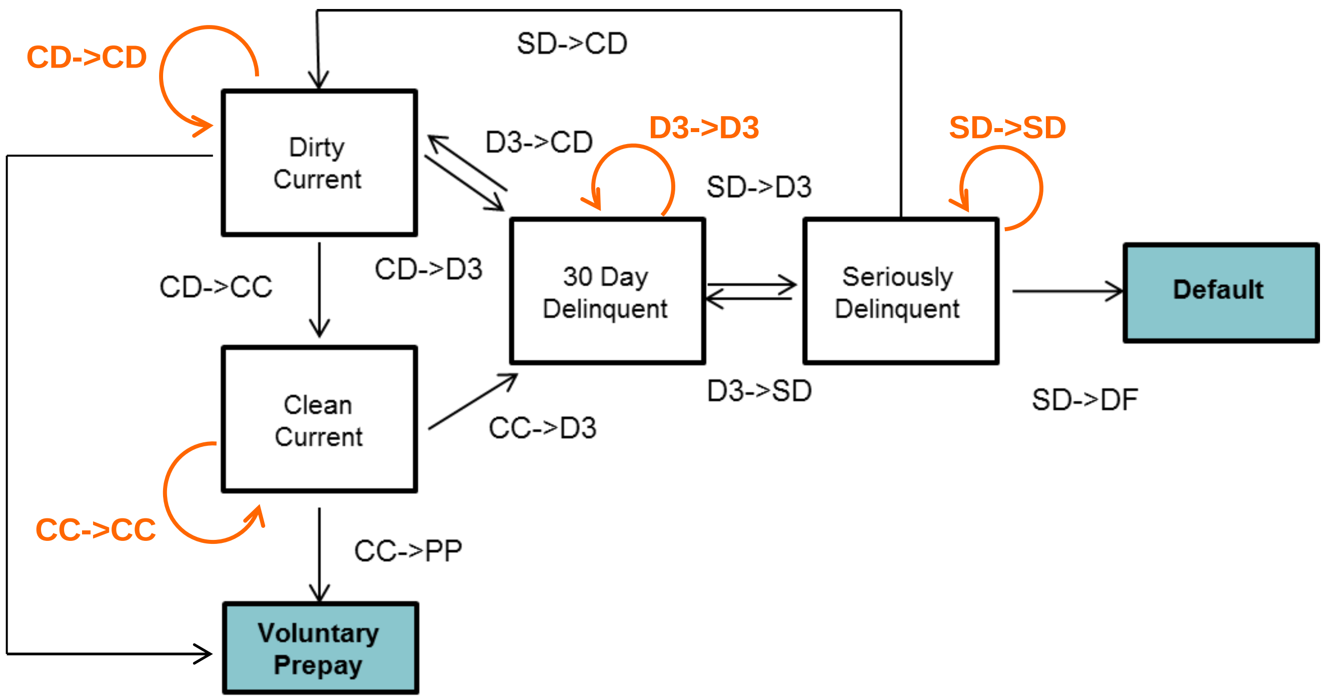
$$S(t) = \begin{bmatrix} \%CC \\ \%CD \\ \%D3 \\ \%SD \\ \%PP \\ \%DF \end{bmatrix}^T \qquad T(t) = \begin{bmatrix} T_{CC\to CC} & 0 & T_{CC\to D3} & 0 & T_{CC\to PP} & 0 \\ T_{CD\to CC} & T_{CD\to CD} & T_{CD\to D3} & 0 & 0 & 0 \\ 0 & T_{D3\to CD} & T_{D3\to D3} & T_{D3\to SD} & 0 & 0 \\ 0 & T_{SD\to CD} & T_{SD\to D3} & T_{SD\to SD} & 0 & T_{SD\to DF} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$S(t+1) = S(t)T(t)$$

$$S(t+n)=S(t)T(t+1)T(t+2)\ldots.T(t+n)$$

Framework of the Roll Rate Model

Graphical illustration of the roll rate model framework

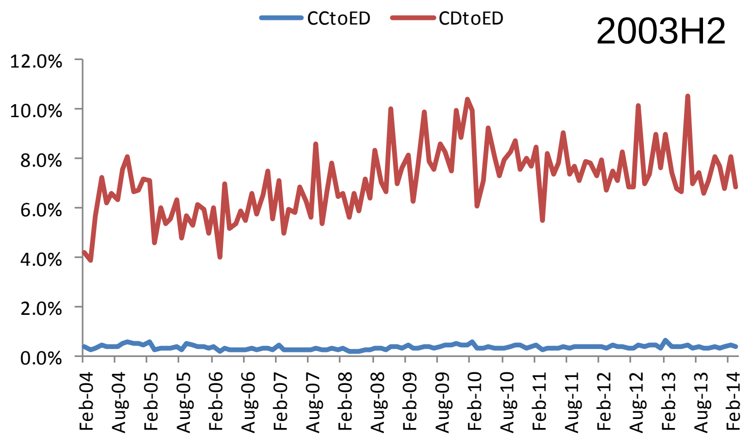


Assumptions:

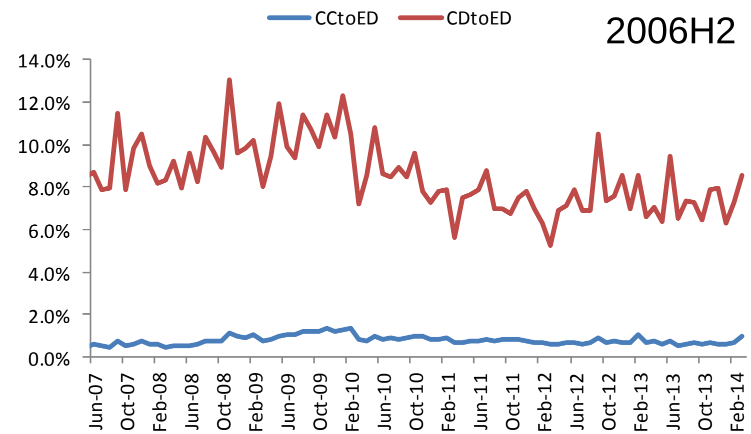
- Voluntary prepayment from Early Delinquency and Seriously Delinquent statuses is assumed to be 0. Delinquent borrowers or borrowers with dirty pay profile, will have impaired credit score making them unlikely to prepay and or to qualify for a new loan.
- Defaults occur from the SD status only.

Dirty Current Roll Rates are Higher versus Clean Current

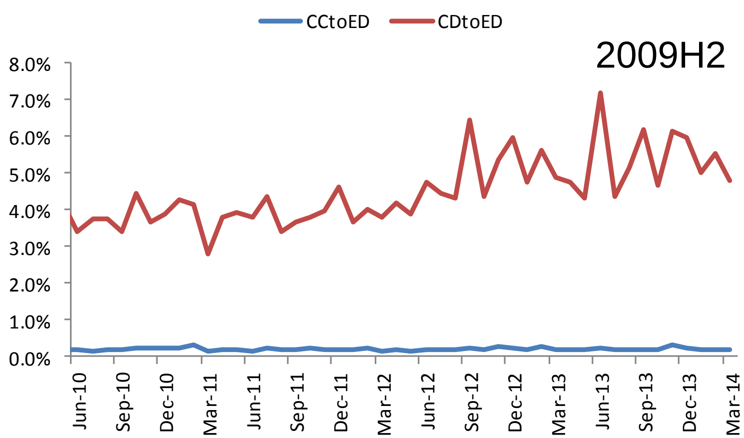
It is important to differentiate between clean current and dirty current borrowers – roll rates of dirty current borrowers are significantly higher even after controlling for initial collateral characteristics. Relative performance of dirty pay borrowers (versus clean pay) has changed for different origination year. Dirty pay new issue loans have a lower roll rate into delinquency, but the rate is relatively high compared to clean pay.



Source: Fannie Mae



Source: Fannie Mae



Source: Fannie Mae

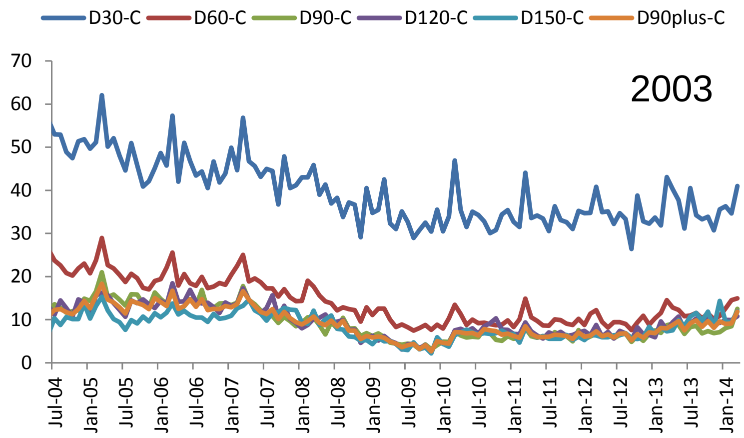
Ratio of roll rate into D3 for clean pay and dirty pay borrowers (70-80 CLTV)

	Multiple During Year		
Orig Year	**2008**	**2010**	**2013**
2003H2	26	22	20
2006H2	13	10	11
2009H2	-	19	24

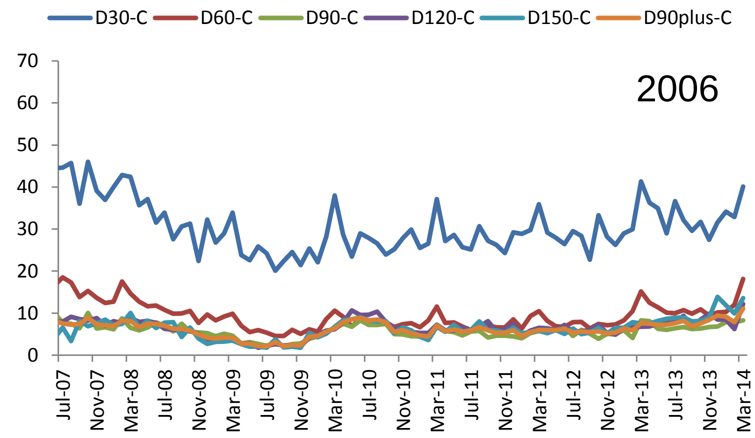
Source: Fannie Mae

Do we Lose Accuracy By Not Segregating D60, D90, D120, and D150?

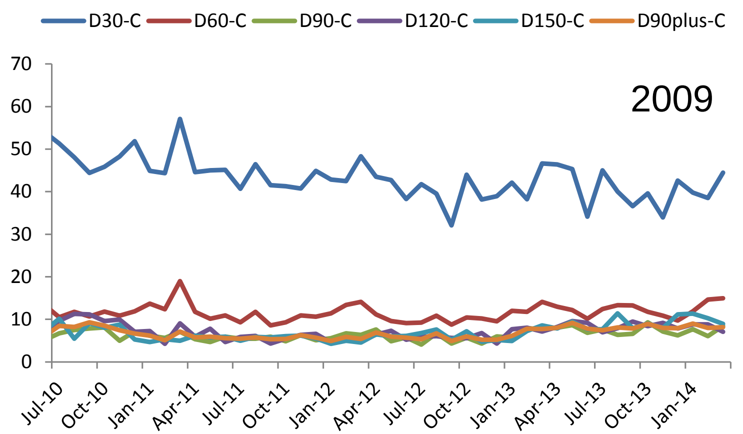
Historical data shows that roll rates from a status that is 90 or more days delinquent is very similar under different scenarios. Therefore, modeling D90, D120, and D150 separately doesn't significantly improve model accuracy.



Source: Fannie Mae



Source: Fannie Mae



Source: Fannie Mae

Models with Fewer Transition States Are More Stable

Curing from D60 is generally higher than from more seriously delinquency statuses, however, these roll rates are correlated. Segregating D60 from higher delinquency statuses mainly impacts short-term projections which can be handled through short-term adjustments.

Having fewer transition states has several advantages:

- Model stability is far better with fewer transition states.
- Small error in any of the transition states can propagate forward, magnifying over time due to the multiplicative nature of the model.
- Allows for better understanding of model projections.
- Computational efficiency is much better, especially for credit-adjusted OAS analysis.

Clean Current to Early Delinquency Roll Rate

We use econometric models to predict various transition roll rates. Clean current to early delinquency roll rates are driven by borrower and loan characteristics, and macroeconomic variables:

- Loan Age (similar to SDA curve)
- FICO
- SATO
- Debt-to-Income (DTI) Ratio
- Occupancy
- Loan Purpose
- HPI Updated CLTV
- Initial down payment (or original CLTV)
- Prepay burnout
- Change in unemployment rate
- HPA momentum (change in home prices over certain period of time)
- Origination Year Adjustment

$$T_{CC\to D3} = f(Loan\ Age) \times f(Borrower\ Score) \times f(CLTV_{adj}) \times f(OCLTV) \times f(HPA\ Momentum) \times f(Factor_{vl}) \times f(Unemp) \times f(Seasonality) \times f(Origination\ Channel)$$

$$f(Borrower\ Score) = F_2(FICO) \times F_3(SATO) \times F_4(Vintage) \times F_5(DTI) \times F_6(\%\ Investor\ Property) \times F_7(Loan\ Purpose)$$

- $Factor_{vl}$ is the factor based on model projected VPRs and captures prepay burnout. Higher prepay burnout implies higher *CC→D3* roll rate everything else being equal.
- $Seasonality$: *CC→D3* roll rates are highest during Nov-Feb period and lowest during Mar-Apr.

Aggregate Fannie Data and Comparison to Performance of Non-Agency Prime Loans

- Cumulative defaults on similar non-agency prime loans are comparable.
- For 2005-2008 origination, prime defaults are higher than Freddie cohort.
- Significant tightening in Freddie underwriting standard after 2008. Average FICO score at origination jumped from 723.4 for 2007 vintage to 761.8 for 2009 vintage. CLTV dropped from 77.1% to 69.1% and DTI dropped from 38% to 33%.

Orig Year	Balance ($B)	% Dist.	CLTV	% CLTV > 80	% Owner	DTI	FTHB (of Purchase)	Purchase	Refi	FICO	% w Att. 2nds	Orig WAC	% CA	% Prepaid	% Defs	% Defs (Freddie)	Defs Prime Non-Ag
1999	16.0	0.5%	79.5	38.3%	94.1%	34.9	18.2%	72.3%	27.7%	709.4	0.5%	7.80	13.9%	92.6%	2.7%	1.2%	1.1%
2000	141.1	4.1%	79.3	37.5%	93.3%	35.7	17.9%	75.7%	24.3%	711.8	1.6%	8.13	14.8%	86.8%	2.3%	1.3%	1.3%
2001	349.9	10.1%	75.8	27.6%	93.5%	34.0	7.7%	37.9%	62.1%	712.9	2.7%	6.99	19.4%	90.0%	2.0%	1.2%	0.7%
2002	374.7	10.8%	73.4	22.7%	92.5%	34.0	6.6%	33.4%	66.6%	717.1	3.9%	6.50	20.6%	78.2%	2.2%	1.4%	0.7%
2003	497.2	14.3%	71.5	19.2%	92.4%	33.5	4.8%	26.0%	74.0%	719.6	5.8%	5.75	22.4%	72.9%	3.1%	2.0%	1.5%
2004	201.0	5.8%	73.7	25.0%	91.7%	36.6	10.6%	43.1%	56.9%	714.9	12.0%	5.84	18.6%	68.1%	5.2%	4.0%	3.0%
2005	208.8	6.0%	73.3	25.8%	91.3%	38.2	9.7%	42.1%	57.9%	719.3	15.9%	5.84	14.6%	62.0%	9.0%	7.2%	7.7%
2006	172.9	5.0%	73.9	27.3%	90.0%	39.3	10.9%	44.5%	55.5%	718.0	16.4%	6.42	12.1%	62.4%	12.2%	10.4%	11.0%
2007	218.6	6.3%	75.5	31.5%	88.9%	39.4	11.2%	41.1%	58.9%	718.4	15.5%	6.36	12.1%	57.6%	13.8%	11.5%	12.2%
2008	263.2	7.6%	75.0	27.8%	88.4%	38.5	14.3%	44.7%	55.3%	739.8	10.8%	6.04	19.8%	61.8%	7.4%	7.5%	13.1%
2009	418.2	12.1%	69.4	12.9%	92.0%	34.2	9.0%	26.1%	73.9%	761.0	10.8%	4.97	23.1%	51.4%	0.8%	0.7%	-
2010	303.9	8.8%	71.0	13.8%	90.2%	32.8	13.1%	36.6%	63.4%	764.3	11.1%	4.72	29.7%	36.1%	0.2%	0.2%	-
2011	243.0	7.0%	72.9	18.9%	87.7%	33.1	13.4%	42.8%	57.2%	762.6	9.7%	4.55	29.7%	21.2%	0.1%	0.0%	-
2012	60.7	1.8%	71.1	17.2%	88.8%	31.9	8.8%	27.2%	72.8%	766.3	10.5%	4.08	32.3%	8.6%	0.0%	-	-

* Includes 30 year prime fixed-rate fully amortizing loans.

Agency defaults include loans with 3rd party sale, short sale, deed-in-lieu of foreclosure, REO acquisition prior to D180, and D180

Non-Agency defaults include all loans paid off from D60, D90, FC, and REO as well as loans that are currently in D90, FC, and REO

Source: Fannie Mae, Freddie Mac, CoreLogic

Cum Defaults for Different Vintage and Buckets – Fannie Data

Data used for calibration:

FICO	2000	2003	2006	2007	2010	2011	2012
501-600	7.7%	9.9%	31.4%	35.4%	8.3%	3.3%	0.6%
601-680	4.0%	6.9%	25.1%	28.4%	2.6%	1.4%	0.4%
681-700	1.7%	4.4%	17.6%	19.9%	1.3%	0.7%	0.2%
701-720	1.2%	3.4%	15.0%	16.9%	0.9%	0.4%	0.1%
721-740	0.7%	2.5%	12.3%	14.0%	0.6%	0.3%	0.1%
741-760	0.5%	1.7%	9.6%	10.9%	0.4%	0.2%	0.0%
761-780	0.3%	1.1%	6.4%	7.4%	0.2%	0.1%	0.0%
781-800	0.3%	0.8%	4.2%	4.9%	0.1%	0.1%	0.0%
>= 800	0.3%	0.8%	3.4%	3.7%	0.1%	0.0%	0.0%

DTI	2000	2003	2006	2007	2010	2011	2012
<= 10	1.6%	1.9%	7.4%	9.0%	0.1%	0.1%	0.0%
11-20	1.4%	1.9%	6.3%	6.9%	0.1%	0.1%	0.0%
21-30	1.5%	2.7%	9.0%	9.3%	0.2%	0.1%	0.0%
31-40	2.0%	3.8%	13.6%	14.9%	0.4%	0.2%	0.1%
41-50	2.2%	4.7%	17.6%	20.3%	0.7%	0.3%	0.1%
51-60	2.0%	4.8%	18.5%	22.2%	1.5%	1.4%	0.0%
61-70	1.9%	4.7%	19.5%	22.9%	0.9%	11.3%	0.0%

CLTV	2000	2003	2006	2007	2010	2011	2012
<= 40	0.7%	0.7%	2.9%	3.2%	0.1%	0.1%	0.0%
41- 50	0.7%	1.1%	5.8%	5.9%	0.1%	0.1%	0.0%
51- 60	0.8%	1.7%	9.4%	9.7%	0.2%	0.1%	0.0%
61- 70	1.3%	2.8%	15.4%	15.1%	0.3%	0.2%	0.0%
71- 80	1.3%	3.7%	15.7%	16.0%	0.4%	0.2%	0.0%
81- 90	2.5%	5.4%	17.6%	22.1%	0.6%	0.3%	0.1%
91-100	3.0%	6.9%	17.3%	22.3%	0.9%	0.4%	0.1%
101-110	5.9%	8.3%	34.2%	31.7%	4.4%	0.0%	0.0%

SATO	2000	2003	2006	2007	2010	2011	2012
<=-0.50	1.0%	3.0%	10.7%	10.2%	0.2%	0.1%	0.0%
-0.49--0.25	1.0%	2.9%	10.8%	10.5%	0.1%	0.1%	0.0%
-0.24- 0.00	1.3%	2.9%	12.2%	13.0%	0.2%	0.1%	0.0%
0.01- 0.25	1.7%	3.5%	16.1%	18.4%	0.3%	0.2%	0.0%
0.26- 0.50	2.6%	4.5%	20.4%	25.9%	0.6%	0.3%	0.1%
> 0.50	5.3%	6.5%	25.3%	32.8%	1.2%	0.6%	0.2%

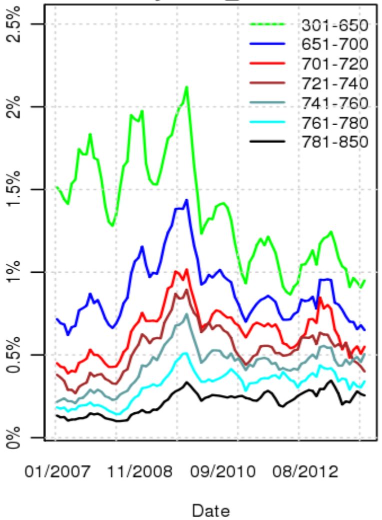
Purpose	2000	2003	2006	2007	2010	2011	2012
Cashout	2.9%	3.6%	18.6%	20.2%	0.7%	0.4%	0.1%
Refinance	3.3%	3.1%	15.1%	20.2%	0.3%	0.1%	0.0%
Purchase	1.5%	4.0%	10.6%	11.1%	0.4%	0.2%	0.1%

Occupancy	2000	2003	2006	2007	2010	2011	2012
Investor	2.6%	3.8%	15.4%	18.9%	0.4%	0.2%	0.0%
Principal	1.9%	3.5%	14.6%	16.6%	0.4%	0.2%	0.0%
Second	0.7%	2.3%	11.6%	11.5%	0.2%	0.1%	0.1%

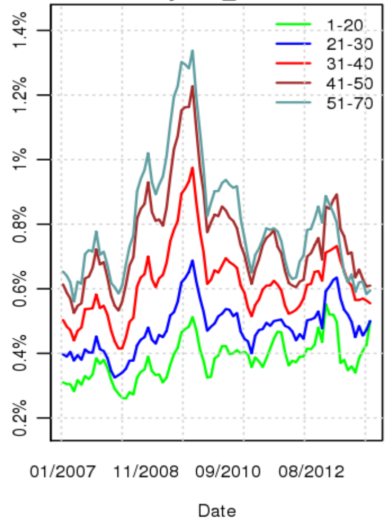
Source: Fannie Mae

Actual Clean Current to Early Delinquency Roll Rate

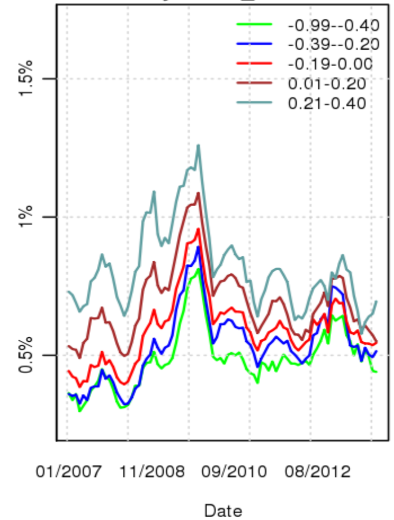
Roll Rate CCtoED
Vintage 2006
By FICO_BKT



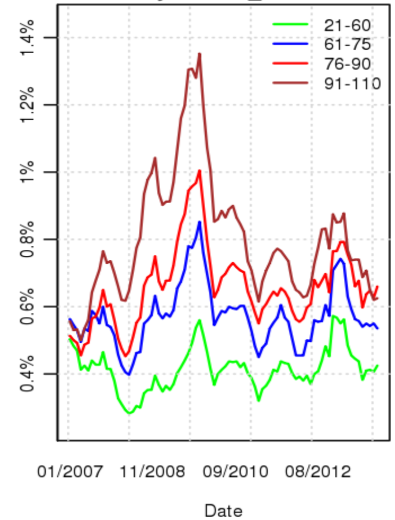
Roll Rate CCtoED
Vintage 2006
By DTI_BKT



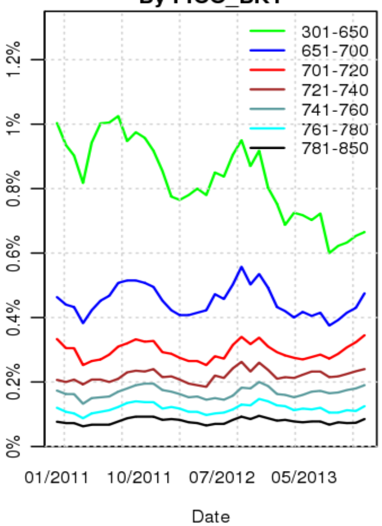
Roll Rate CCtoED
Vintage 2006
By SATO_BKT



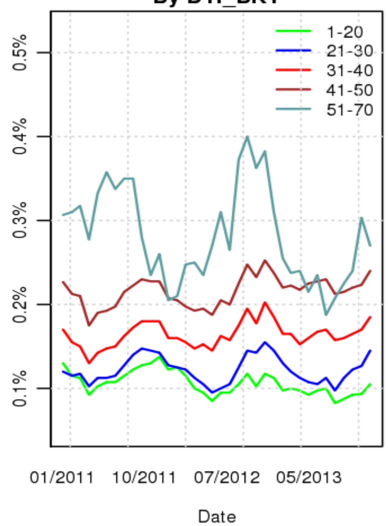
Roll Rate CCtoED
Vintage 2006
By CCLTV_BKT



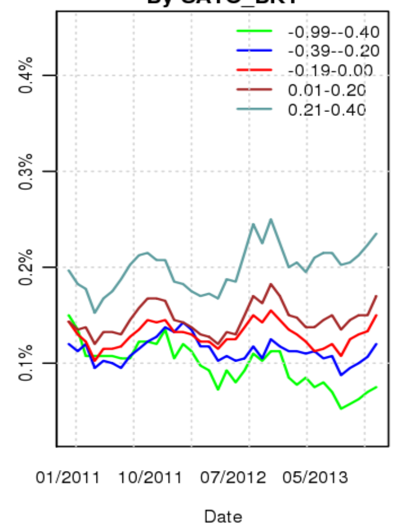
Roll Rate CCtoED
Vintage 2010
By FICO_BKT



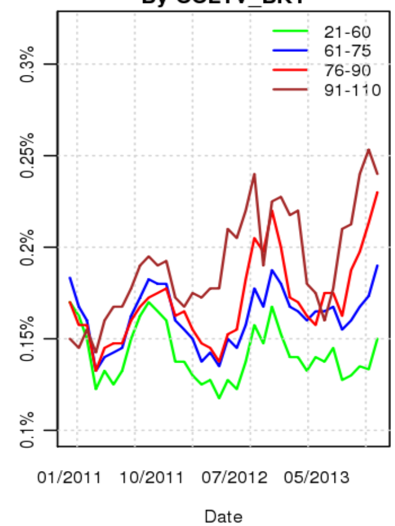
Roll Rate CCtoED
Vintage 2010
By DTI_BKT



Roll Rate CCtoED
Vintage 2010
By SATO_BKT

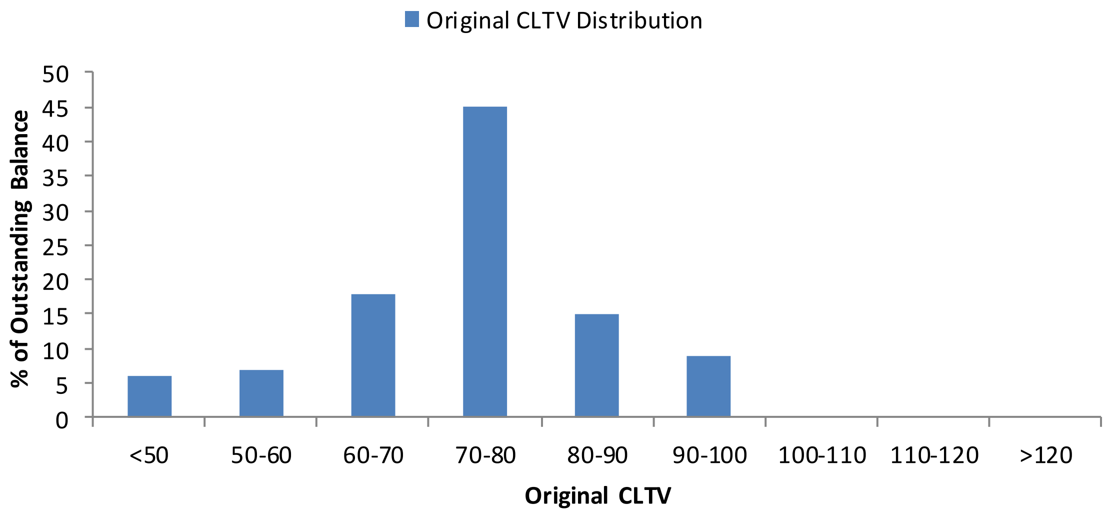


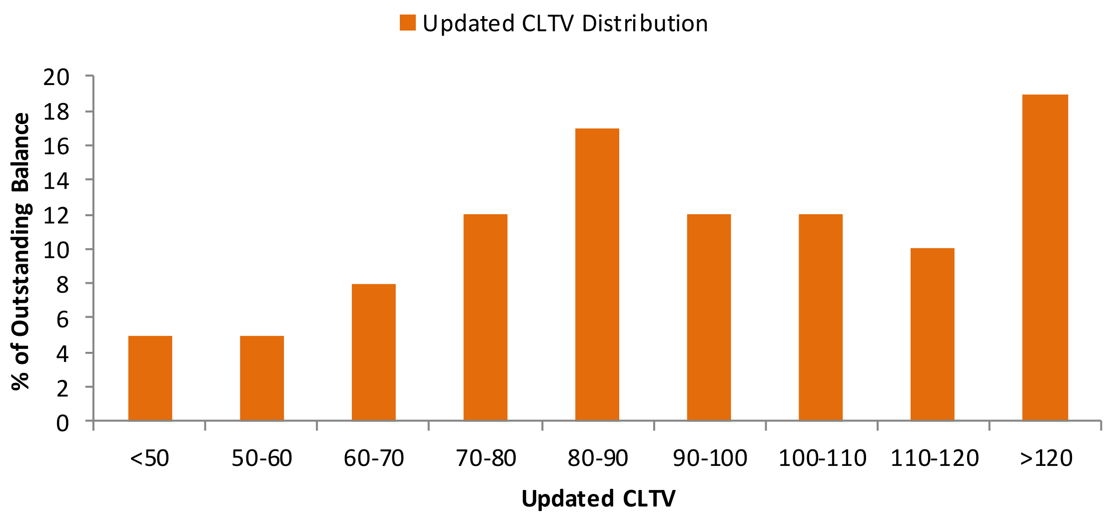
Roll Rate CCtoED
Vintage 2010
By CCLTV_BKT



Source: Fannie Mae

Model Mechanics Illustration – CLTV Update



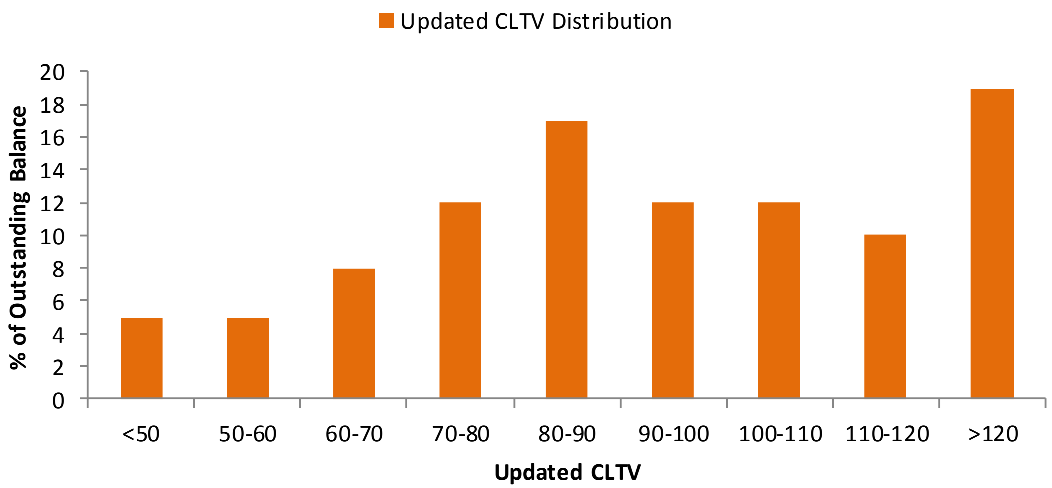


Source: Citi

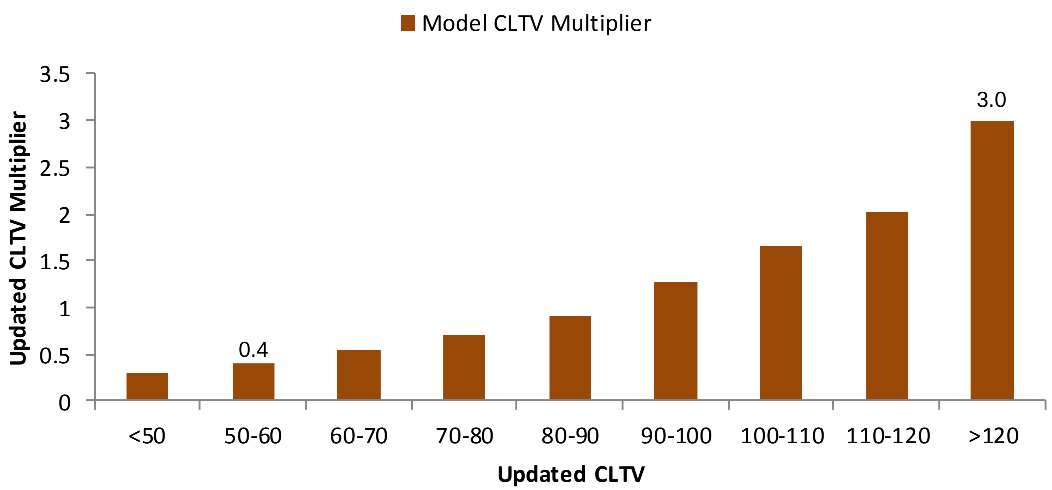
Original CLTV of the cohorts are input to the model.

The model calculates updated CLTVs based on actual historical HPI and projected future HPA.

Model Mechanics Illustration – CLTV Multiplier



Model projects defaults based on various collateral inputs including current Combined LTV. Higher CLTV results in higher multiplier causing projected defaults to increase.



Source: Citi

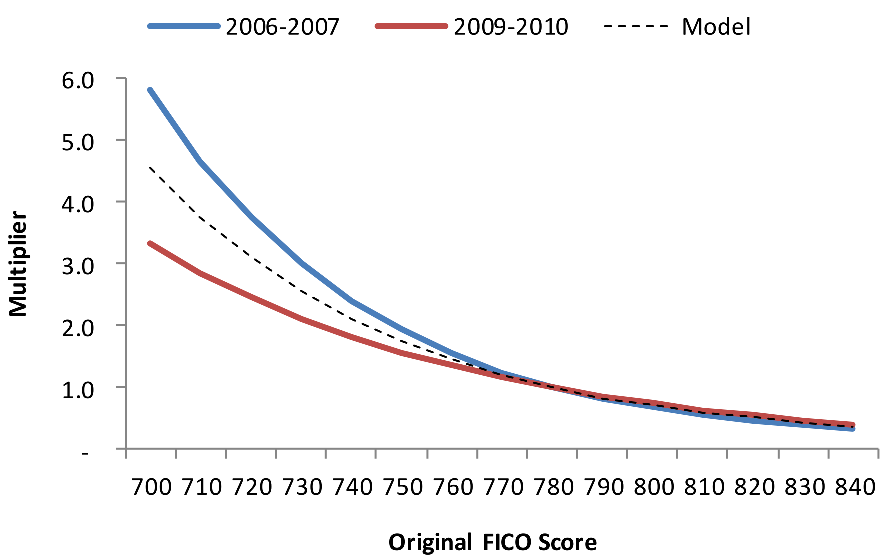
Initial estimates are typically derived using logistic regressions or non-parametric regression where shape of the function is not known. **Need to consider correlation, especially for non-parametric regression**.

Are Borrower Risk Measures Consistent Across Time?

Everything being equal, is creditworthiness of a borrower with 720 FICO from 2006 vintage the same as another borrower with same FICO from 2010 vintage? Regression of historical data shows that riskiness of a borrower as measured by FICO has changed over time.

- Prior to the housing crisis, whenever a borrower got into trouble, he was able to sell the house and payoff the loan in full. This prevented his FICO from being depressed. FICO scores of borrowers were also boosted as they were able to stay current for a longer period of time by refinancing into another affordability product.
- Borrowers with post-crisis high FICO were positively self-selected – they were able to stay current even during the housing downturn.

Parameter estimates for FICO for different origination year



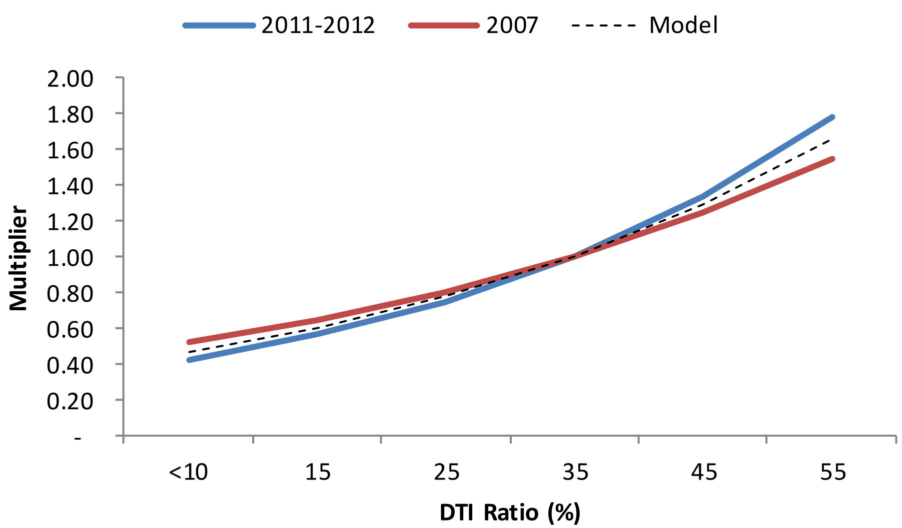
Source: Citi

Inflation in FICO score seemed to have occurred more for the lower FICO range (<740). Credit worthiness as measured by FICO did not change much for borrowers with higher credit score.

DTIs for 2006-2007 Origination Year Were Not Significantly Understated

For 2006 vintage non-agency loans, particularly for subprime, we found that debt-to-income (DTI) ratio showed very little correlation to default rates. This clearly indicated fraudulent underwriting where for many loans actual DTIs were higher than reported values. We suspect, that there was some fraud in reported DTIs for agency loans as well, although to a much lesser extent.

Parameter estimates for DTI for different origination year



Source: Citi

Differences in Underwriting Standards are Captured by Origination Year Adjustments

Adjustment for origination year can be based on average original loan characteristics and through fitting process. Lax underwriting is generally reflected in average characteristics of loans originated during that period. During the housing boom, a significant percentage of loans were originated with high DTIs. A significant percent were affordability products (IOs, Negam, etc.)

Orig Yr	FICO	CLTV	DTI	%With Att. 2nds	% CLTV > 80	% DTI > 40	% Cashout
2003	720	71.55	33.5	6%	19%	29%	32%
2004	715	73.74	36.6	12%	25%	38%	32%
2005	719	73.43	38.2	16%	26%	43%	41%
2006	718	73.98	39.3	17%	27%	47%	40%
2007	718	75.51	39.4	16%	32%	47%	39%
2008	740	75.03	38.5	11%	28%	45%	29%
2009	761	69.4	34.2	11%	13%	31%	28%
2010	764	71.01	32.8	11%	14%	25%	21%
2011	763	72.93	33.1	10%	19%	26%	18%
2012	765	72.86	31.9	10%	21%	22%	16%
2013	759	75.39	32.9	9%	28%	25%	16%

Source: Fannie Mae

How Much of "Additional" Risk Does SATO Capture?

SATO is often used as a measure of "additional" risk not observable in the reported data. During periods of tighter underwriting, reported data (FICO, DTI, etc) is more reliable and SATO does not capture significant additional risk that is not already observed in the data. As a result SATO shows much higher correlation to loan characteristics during periods of tighter underwriting.

WAC for Different FICO and DTI Buckets

	FICO							
Orig Qtr	**680**	**700**	**720**	**740**	**760**	**780**	**800**	**820**
2006Q3	6.65	6.64	6.63	6.62	6.60	6.58	6.57	6.57
2007Q3	6.64	6.62	6.61	6.60	6.59	6.56	6.54	6.54
2010Q3	4.81	4.80	4.72	4.65	4.63	4.62	4.60	4.58

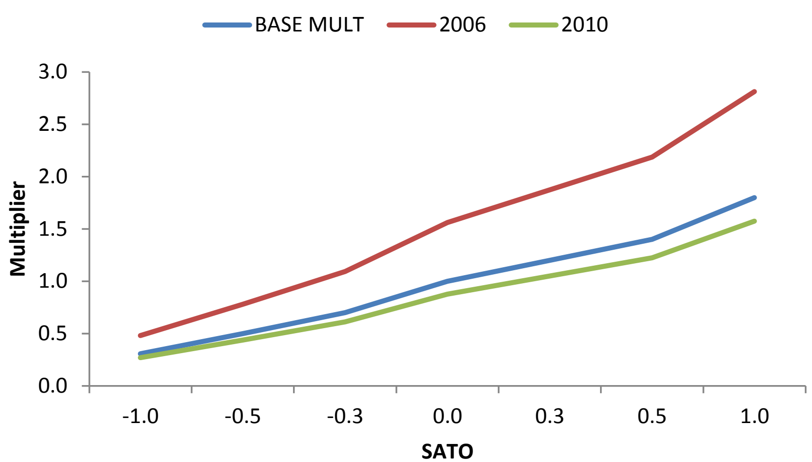
	DTI				
Orig Qtr	**30**	**40**	**50**	**60**	**70**
2006Q3	6.58	6.60	6.61	6.62	6.64
2007Q3	6.55	6.57	6.60	6.62	6.62
2010Q3	4.60	4.64	4.66	4.81	4.87

	680	**700**	**720**	**740**	**760**	**780**	**800**	**820**
2006Q3	0.03	0.02	0.02	0.00	-0.01	-0.03	-0.05	-0.04
2007Q3	0.04	0.02	0.01	0.00	-0.02	-0.04	-0.06	-0.06
2010Q3	0.16	0.14	0.07	0.00	-0.02	-0.04	-0.05	-0.07

Orig Qtr	**30**	**40**	**50**	**60**	**70**
2006Q3	0.00	0.02	0.04	0.05	0.06
2007Q3	0.00	0.03	0.05	0.07	0.07
2010Q3	0.00	0.04	0.06	0.20	0.27

Parameter estimates for SATO for different origination year

Source: Fannie Mae



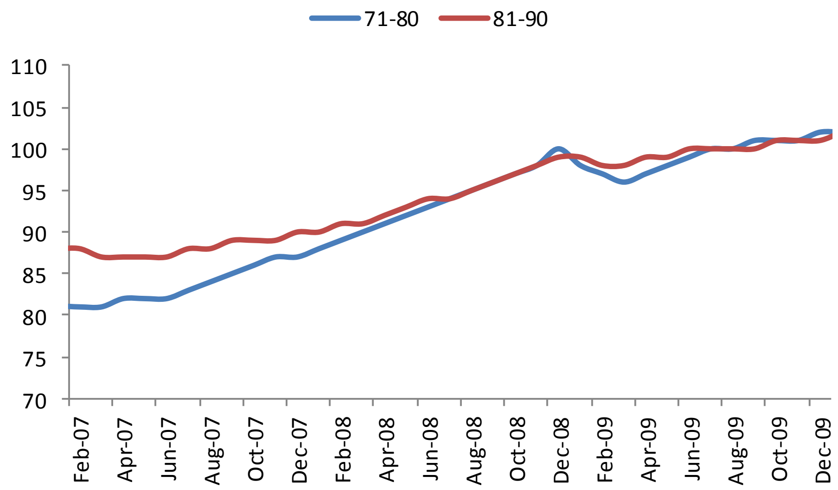
Source: Citi

The model should, therefore, use a flatter SATO multiplier during periods of tighter underwriting standards (since most of the risk is already captured in reported data) and steeper function during periods of lax underwriting.

Loans with Bigger Down Payment have Lower Default Rate

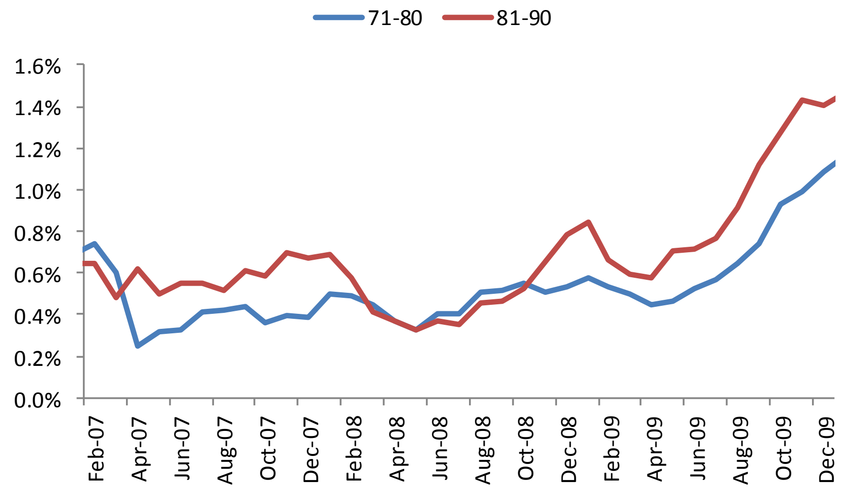
Borrowers who made a bigger down payment on their mortgage, tend to have a lower roll rate into delinquency compared to borrowers who have the same current CLTV but made a lower down payment. The graph below shows updated CLTVs for two original CLTV cohorts and roll rates (2006 origination).

HPI Updated CLTV



Source: Fannie Mae, CoreLogic

CC-D30 Roll Rate

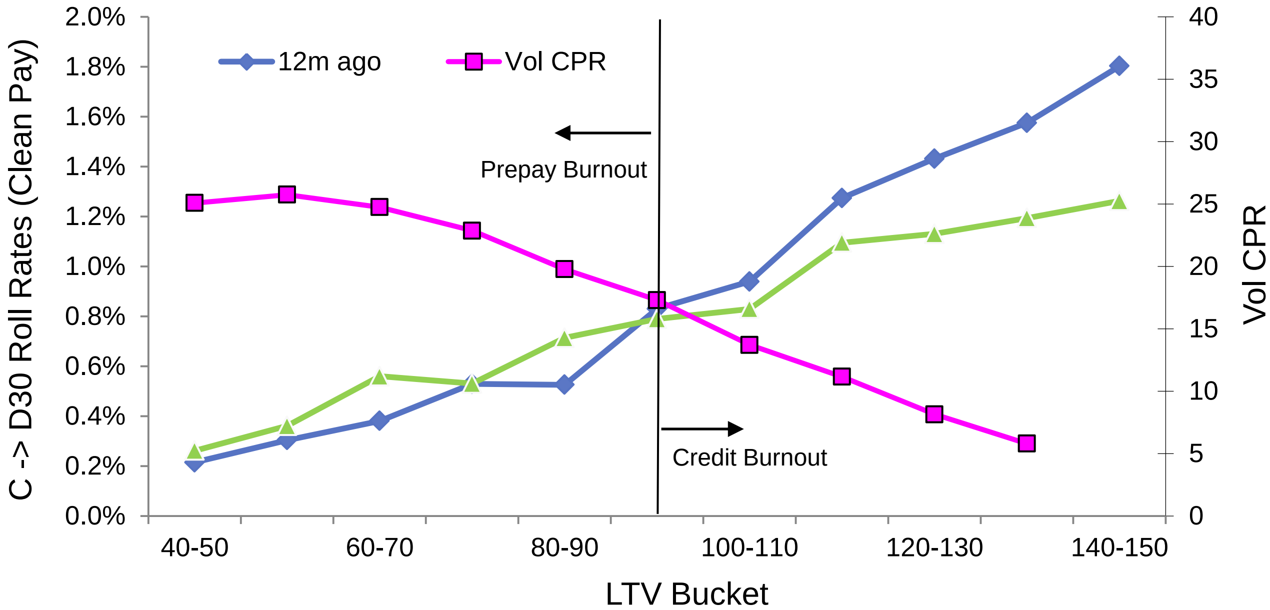


Source: Fannie Mae

Prepay and Credit Burnout

High prepayment results in better borrowers leaving the pool thus causing credit risk to increase. Prepay burnout results in worsening borrower and loan characteristics which is captured when the collateral is run at loan or cohort level (the weight a cohort gets reduces if the balance of that cohort drops due to voluntary prepayment). However, there is burnout effect even in cohorts consisting of very similar loans. This is captured in the model by estimating the current factor based on model projected historical voluntary CPR: $f(Factor_{vl})$. The lower the factor, the higher is the multiplier.

Non-Agency Jumbo Prime (As of July 2010)



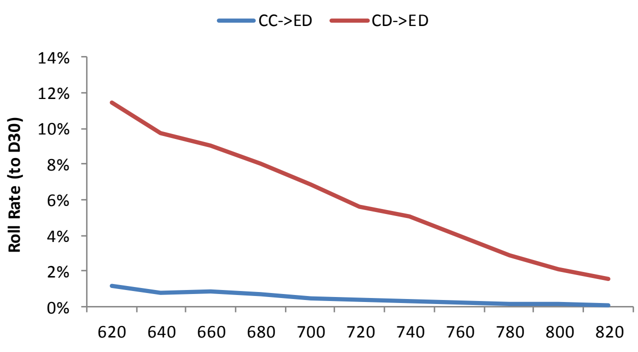
Source: CoreLogic

Dirty Current to Early Delinquency Roll Rate

Most of the drivers of CD-D30 roll rate are the same as those for clean pay borrowers, although the sensitivity to these factors are quite different. For example, while original FICO is still a predictor of CD-D30 roll rate, the multiplier function is much flatter than for CC-D30 roll rate. Similarly DTI function is flatter - borrowers with low original DTI who become delinquent may have some financial stress (due to other debt) and their DTI may have gone up. However they are less likely to become delinquent if they cure compared to a dirty pay borrower with similar current DTI as their financial stress may have been temporary.

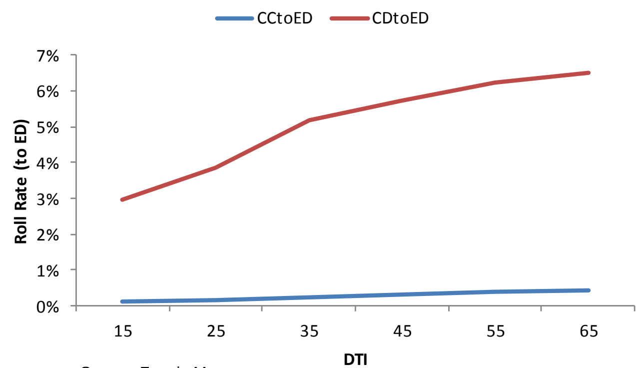
$$T_{CD\to D3} = F_{23}(AGE) \times F_{24}(Current\ LTV) \times F_{25}(FICO) \times F_{26}(SATO) \times F_{27}(UNEMP) \times F_{28}(DTI) \times F_{29}(HPA\ Momentum) \times F_{30}(Factor_{vl})$$

Roll Rate as of Dec-2012 (2009 Origination Year)

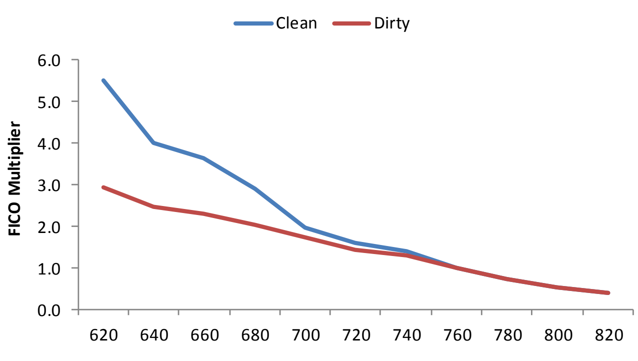


Source: Fannie Mae

Actual Roll Rate

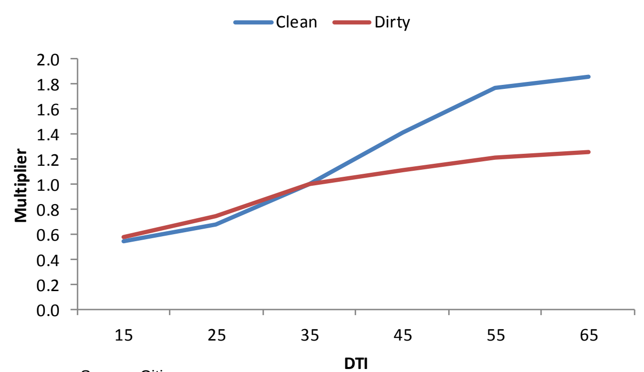


Source: Fannie Mae



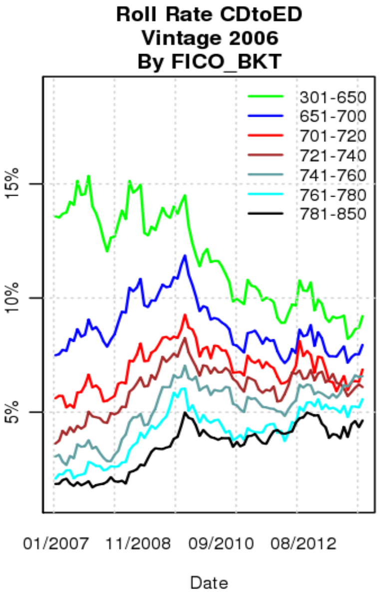
Source: Citi

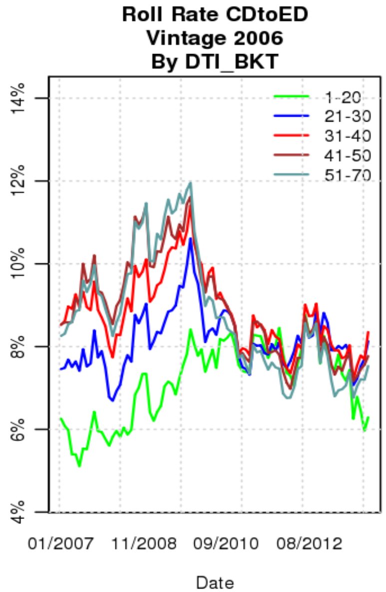
Implied Multiplier

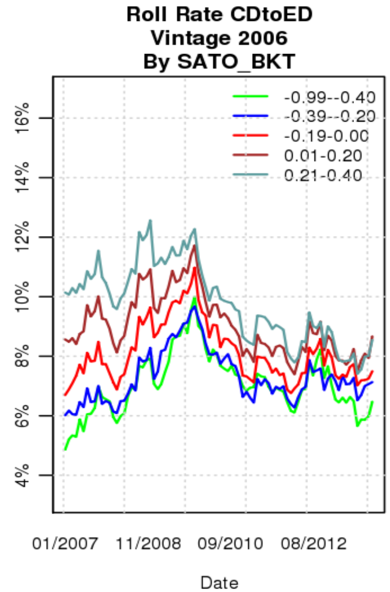


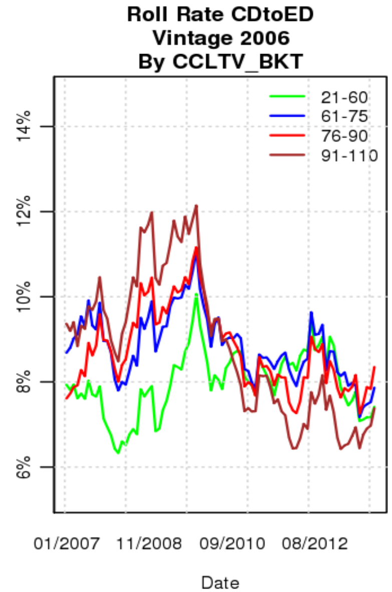
Source: Citi

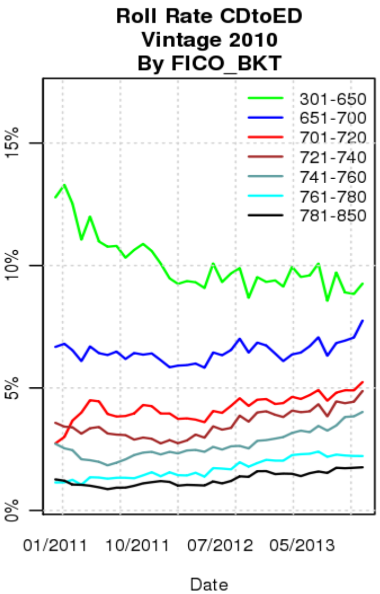
Actual Dirty Current to Early Delinquency Roll Rate

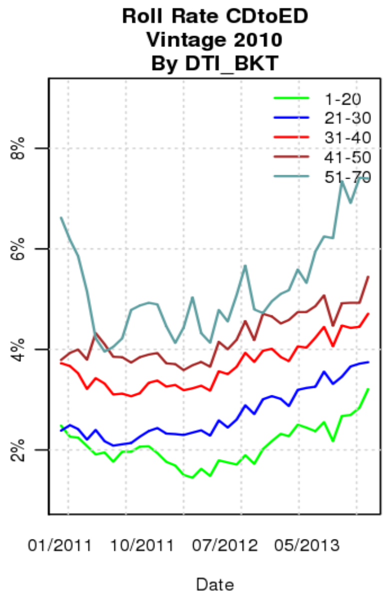


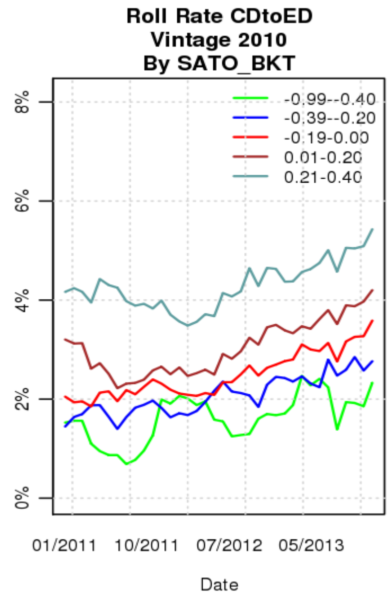


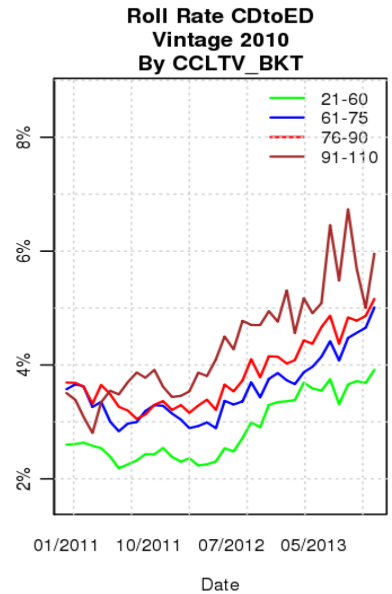












Source: Fannie Mae

Curing from D30 and Seriously Delinquency Status

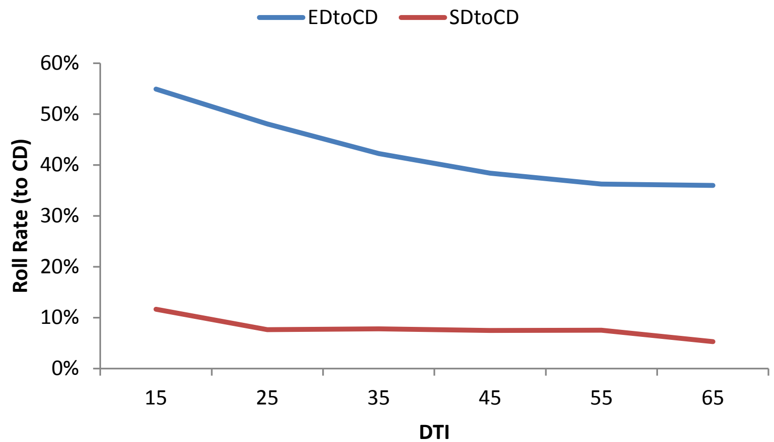
Cure rate from both early (ED-CD) and seriously delinquency (SD-to-better) use the following functional form:

$$T_{D3/SD \to CD} = F_{36}(AGE) \times F_{37}(Current\ LTV) \times F_{38}(FICO) \times F_{39}(SATO) \times F_{40}(UNEMP) \times F_{41}(DTI)$$

Many of the high FICO and low DTI borrowers may miss one payment due to forgetfulness rather than due to financial stress and are likely to cure in the next payment cycle. As a result, cure rates from early delinquency shows very high sensitivity to these variables.

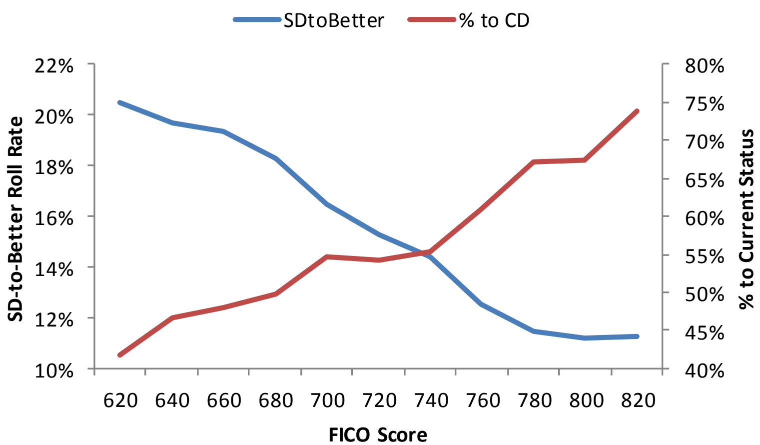
Curing from seriously delinquency status shows little sensitivity to original DTI. High FICO borrowers who cure from SD are more likely to transition to CD than ED compared to low FICO borrowers. Curing from SD also slows with prepay burnout.

Cure Rate from SD and CD



Source: Fannie Mae

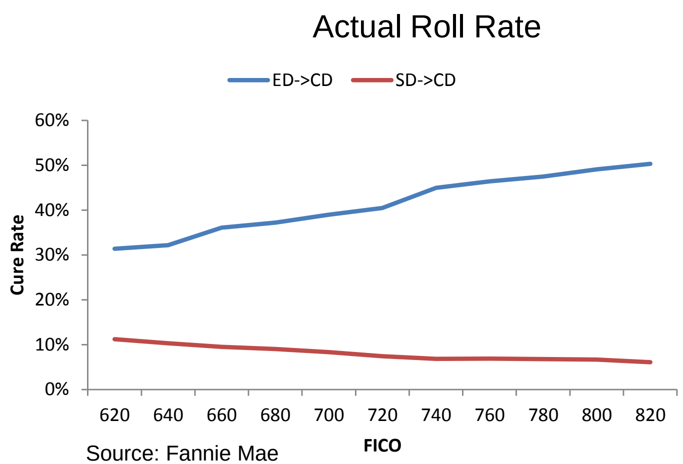
SD-to-Better Transition Rate



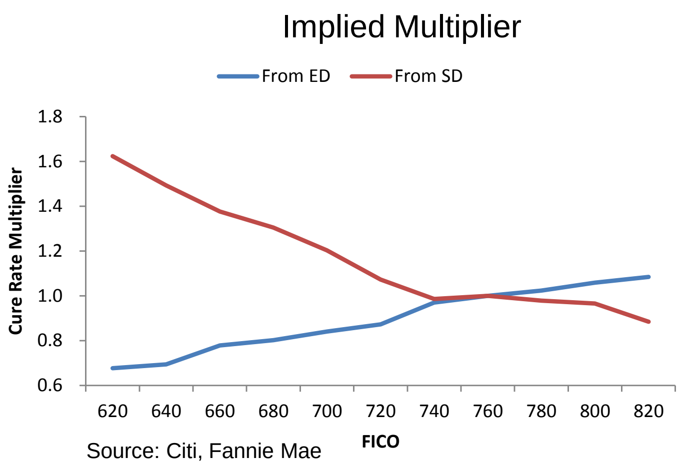
Source: Fannie Mae

Curing from Seriously Delinquency Status

Cure rate from seriously delinquency status is much lower than from early delinquency status. High FICO borrowers who miss two or more payments typically do so when the stress is lasting (loss of job or illness) and, therefore, are less likely to cure even when compared to lower FICO borrowers who in general have a higher tendency to make irregular monthly scheduled payments. Main drivers of cure rate from seriously delinquency seem to be FICO (reversed) and Current CLTV.

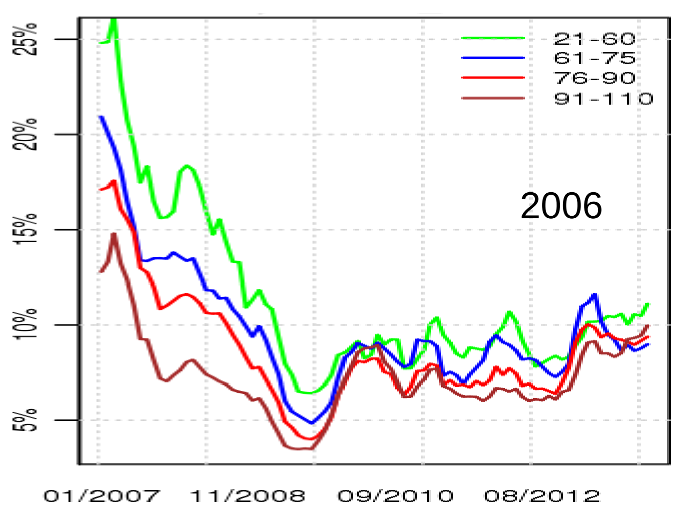


Source: Fannie Mae

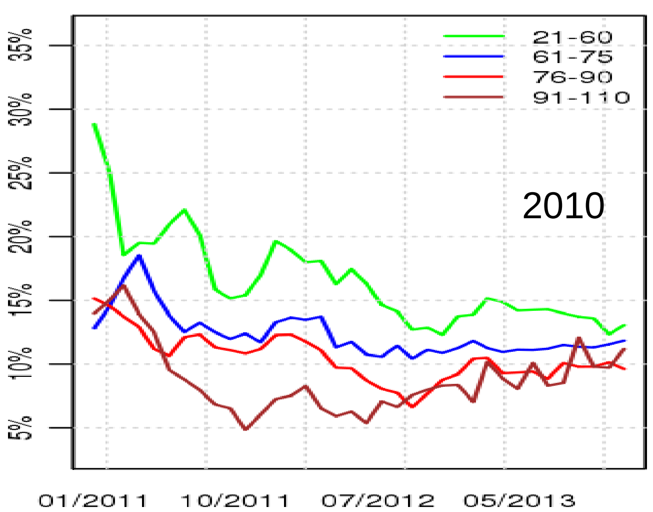


Source: Citi, Fannie Mae

Cure Rate from SD

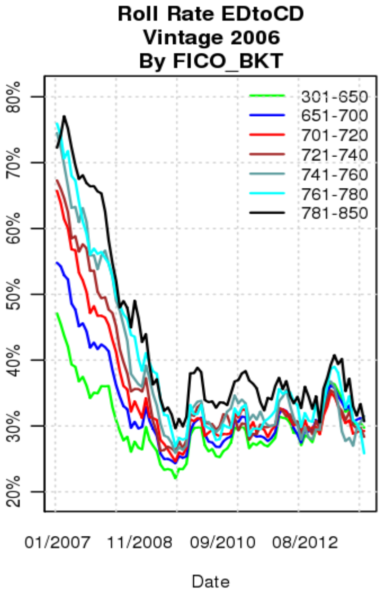


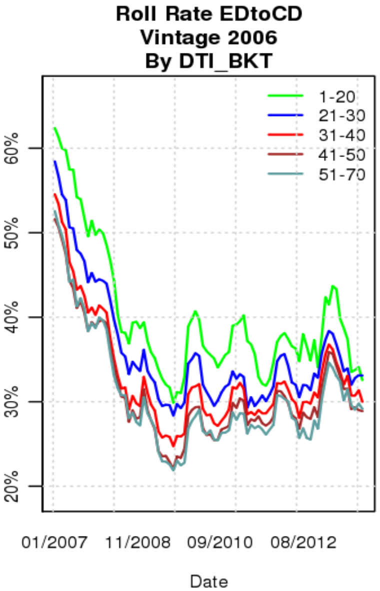
Source: Fannie Mae

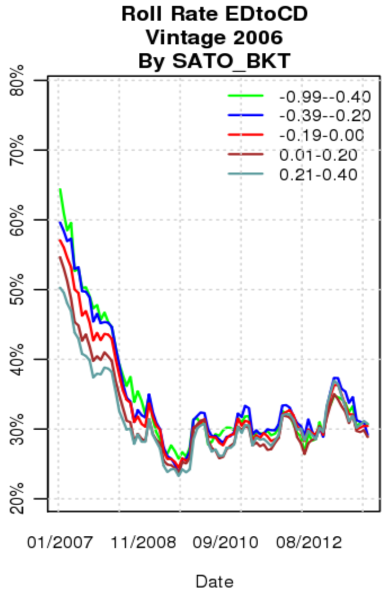


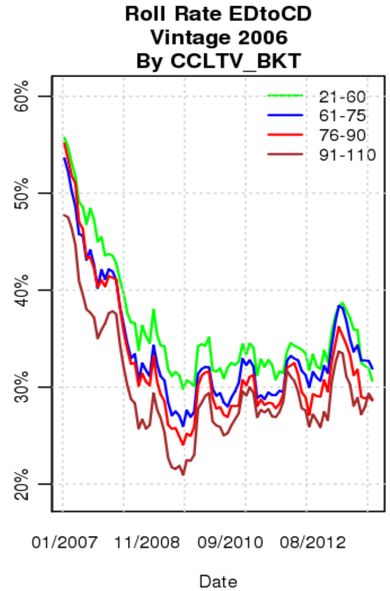
Source: Citi, Fannie Mae

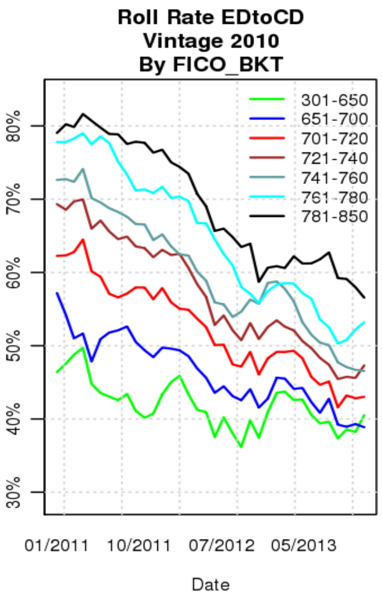
Actual Cure Rates From Early Delinquency

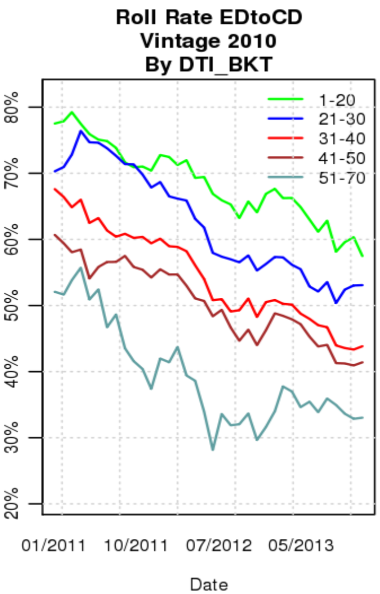


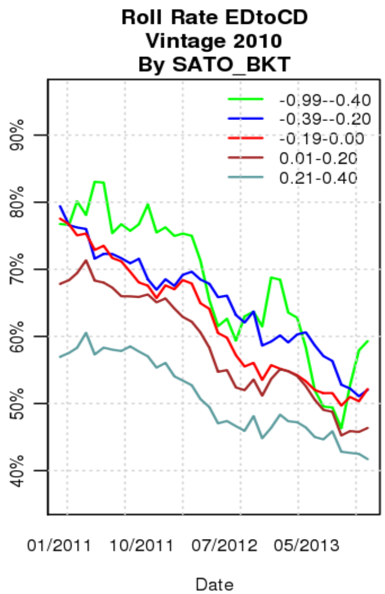


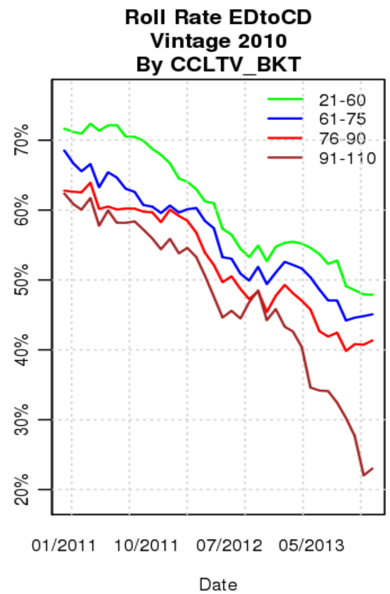










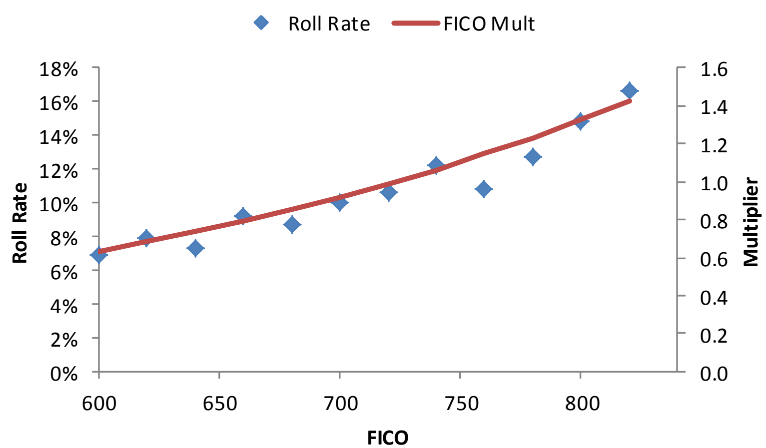


Source: Fannie Mae

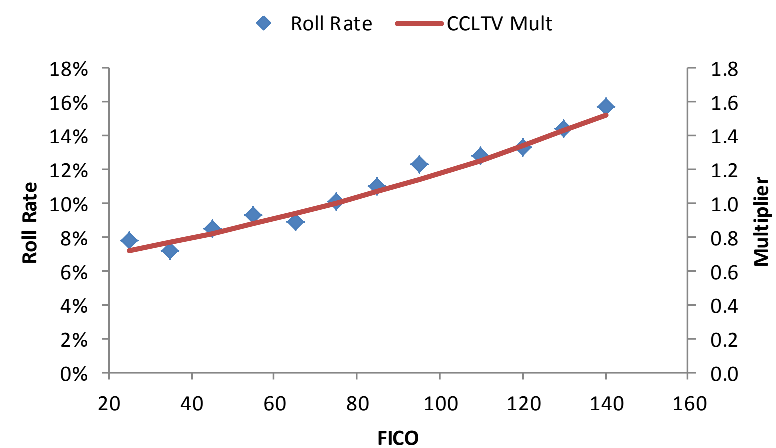
Other Roll Rates

Seriously delinquent to default (SD-DF) roll rate shows little sensitivity to most variables and is modeled using a simple function. Borrowers with high FICO score are more likely to default once they become seriously delinquent. For liquidations, there are additional parameters for states (judicial vs. non-judicial).

$$T_{SD\rightarrow DF} = F_{20}(Age) \times F_{21}(FICO) \times F_{22}(Current\ CLTV) \times F_{23}(Factor_{vl}) \times F_{24}(HPA\ Momentum) \times\ F_{24}(Unemp)$$



Source: Fannie Mae, Citi



Source: Fannie Mae, Citi

Roll rate from early delinquency to seriously delinquency is modeled as follows:

$$T_{D3\rightarrow SD} = f(Loan\ Age) \times f(FICO) \times f(Current\ CLTV) \times f(SATO) \times f(HPA\ Momentum) \times f(DTI) \times\ f(Unemp) \times f(Vintage) \times f(Seasonality)$$

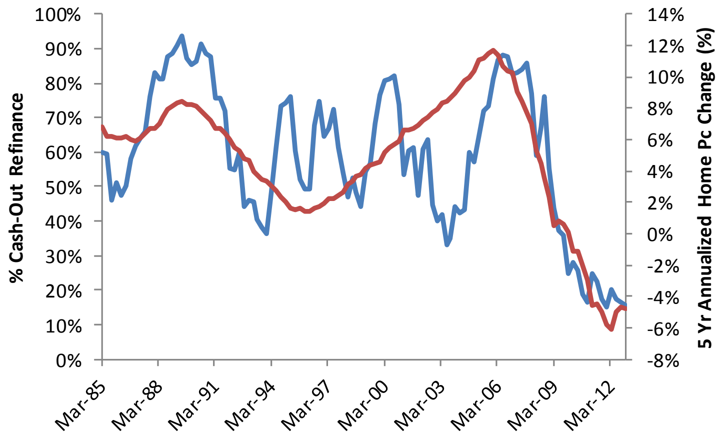
Prepayment Model Consideration

Refinancing patterns will change because of regulatory, market, or borrower changes:

- Incorporate lower refinance rate due to tighter underwriting and higher refinance rate during periods of lax underwriting. This is handled by an additional spread based on borrowers ability to refinance into one of the several "eligibility buckets".

$$\mathrm{MRate}' = \mathrm{MRate} + \mathrm{SPRD}_{\mathrm{elig}}$$

- During the housing boom period of 2004-2007, a significant percentage of the refinance were cash-out refinancing – some even refinancing to higher mortgage rates (into more affordable products). Model also incorporates flipping.



Source: Fannie Mae, CoreLogic

$$\begin{aligned}\text{Cashout Refi CPR}\\ &= S_1(\mathrm{WAC} - \mathrm{MRate}' + \mathrm{Elbow})\\ &* F_1(\mathrm{CombinedLTV}) * F_2(\mathrm{HPA})\\ &* F_3(\mathrm{HPA\ Momentum}) * \cdots * F_7(\mathrm{FICO})\\ &* F_8(\mathrm{LoanSzRatio}) * S_2(\mathrm{Burnout})\end{aligned}$$

- Dirty pay borrowers who maintain a clean pay history become eligible to refinance.

Model Projections: CAS 2014-C04

	HPA	Unemp
Opt	+5, +5, +5, +3	4.5
Base	Flat +3	Flat 5.5
Sev 1	-5, -3, -2, +3	7, 6, 5.5
Sev 2	-10, -5, -5, +3	8, 7, 6, 5.5

	Orig C/E (%)	
	Group1	Group 2
AH	3.00	3.75
M1	2.00	2.55
M2	0.30	0.65
BH	0.00	0.00

CAS 2014-C04 (Group 1)

Tranche	Px (2-Feb)	Scenario	Yield	DM	WAL	PWD	Cum Defs
M1	**100.22**	**Opt**	**2.23**	**175**	**1.09**	**0.0**	**1.42**
		Base	2.26	175	1.10	0.0	1.59
		Sev 1	2.36	178	1.27	0.0	2.37
		Sev 2	2.44	179	1.43	0.0	2.97
M2	**101.81**	**Opt**	**6.09**	**455**	**6.02**	**0.0**	**1.42**
		Base	6.12	455	6.09	0.0	1.59
		Sev 1	5.42	381	6.94	7.4	2.37
		Sev 2	3.82	220	7.39	21.3	2.97

CAS 2014-C02 (Group 2)

Tranche	Px (2-Feb)	Scenario	Yield	DM	WAL	PWD	Cum Defs
M1	**100.56**	**Opt**	**2.22**	**168**	**1.29**	**0.0**	**2.38**
		Base	2.28	168	1.33	0.0	2.71
		Sev 1	2.54	178	1.76	0.0	4.08
		Sev 2	2.76	184	2.20	0.0	5.03
M2	**101.70**	**Opt**	**6.19**	**465**	**5.93**	**0.0**	**2.38**
		Base	6.26	468	6.21	0.0	2.71
		Sev 1	6.03	435	7.78	4.2	4.08
		Sev 2	4.77	306	8.42	17.5	5.03

Collat Distribution

OLTV	Group 1	Group 2
61-70	19.3	-
71-80	74.2	-
81-90	5.3	40.9
91-100	1.3	59.1

FICO	Group 1	Group 2
601-650	1.5	0.6
651-700	9.5	10.0
701-750	25.5	32.3
751-800	51.1	49.8
801-850	12.3	7.3

DTI	Group 1	Group 2
1-20	9.6	4.7
21-30	26.8	25.8
31-40	35.1	42.0
41-80	28.6	27.5

Source: Fannie Mae, Citi

Model Projections: STACR 2015-DN1

	HPA	Unemp
Opt	+5, +5, +5, +3	4.5
Base	Flat +3	Flat 5.5
Sev 1	-5, -3, -2, +3	7, 6, 5.5
Sev 2	-10, -5, -5, +3	8, 7, 6, 5.5

	Orig C/E (%)
AH	4.50
M1	3.50
M2	2.50
M3	1.00
B	0.00

Tranche	Px	Scenario	Yield	DM	WAL	PWD	Cum Defs
M1	**100.14**	**Opt**	**1.30**	**103**	**0.63**	**0.0**	**1.62**
		Base	1.36	106	0.73	0.0	1.80
		Sev 1	1.39	107	0.78	0.0	2.65
		Sev 2	1.41	108	0.82	0.0	3.24
M2	**101.00**	**Opt**	**2.26**	**172**	**1.48**	**0.0**	**1.62**
		Base	2.37	178	1.63	0.0	1.80
		Sev 1	2.53	186	1.87	0.0	2.65
		Sev 2	2.65	191	2.07	0.0	3.24
M3	**102.11**	**Opt**	**4.84**	**358**	**4.04**	**0.0**	**1.62**
		Base	4.92	361	4.32	0.0	1.80
		Sev 1	5.26	373	5.86	0.0	2.65
		Sev 2	5.58	383	8.28	0.0	3.24
B	**100.00**	**Opt**	**10.26**	**844**	**8.54**	**30.2**	**1.62**
		Base	9.73	793	8.47	34.8	1.80
		Sev 1	5.40	376	7.45	65.3	2.65
		Sev 2	-0.57	-207	6.47	88.8	3.24

Collat Distribution

OLTV	
61-70	16.8
71-80	77.4
81-90	4.9
91-100	0.9

FICO	
601-650	1.8
651-700	11.7
701-750	28.0
751-800	47.3
801-850	11.2

DTI	
1-20	8.0
21-30	24.0
31-40	36.4
41-80	31.6

Source: Fannie Mae, Citi

Stochastic Home Price Models and Credit-Adjusted OASs

- For evaluating prepayment risk, we combine term structure models with prepayment models => OAS analysis;
- Term structure models are ***not*** predictive models of interest rates. They are stochastic models that generate interest rate paths which resemble historical interest rate movements and which are consistent with market prices;
- We present here a corresponding stochastic model for HPA rates. It is not a predictive model, but does generate HPA paths that are consistent with historical behavior;
- Main difficulty is the lack of purely HPA-driven market instruments with which to calibrate the model;
- The stochastic HPA model is combined with a 2-factor term structure model to get cash flows adjusted for both prepay risk and credit risk => credit-adjusted OAS.

A Stochastic HPA Model

Key Assumptions:

- Changes in home prices are correlated with those in previous months;
- Changes in mortgage rates influence home prices, but the effect dissipates over time;

Model: Home price appreciation in month t, h(t), is:

$$h(t) - \mu(t) = a\,[h(t-1) - \mu(t-1)] + \varrho\,[MRATIO(t) - 1] + \sigma\, Z(t)$$

where

$\mu(t)$: The projected equilibrium HPA for month t;

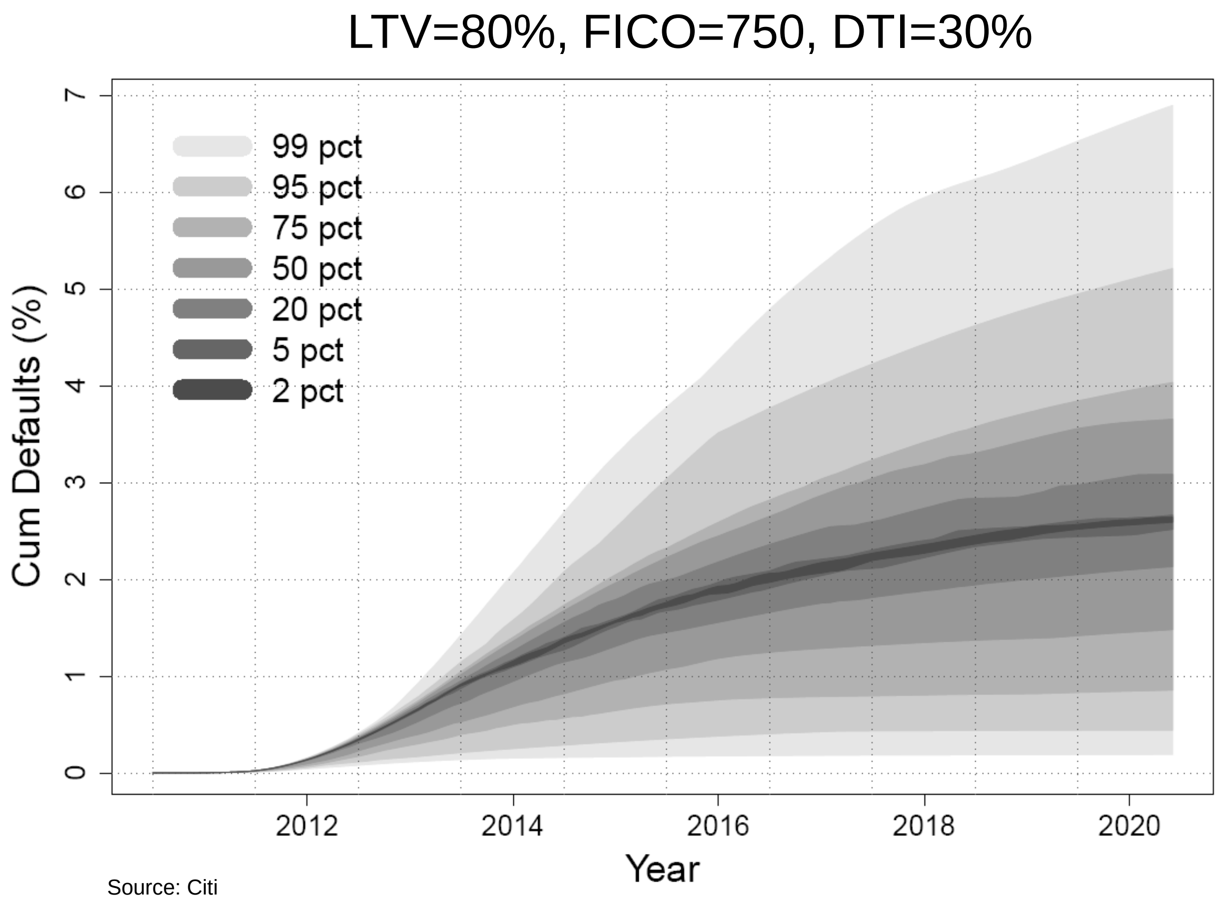
$MRATIO(t)$: Average of past mortgage rates/current mortgage rate;

σ: Volatility of home prices

$Z(t)$: Random shock

Calibration: The $\mu(t)$ can be calibrated to the C-S or Radar Logic (RPX) home price futures. The parameters a, ϱ and σ are estimated using historical data.

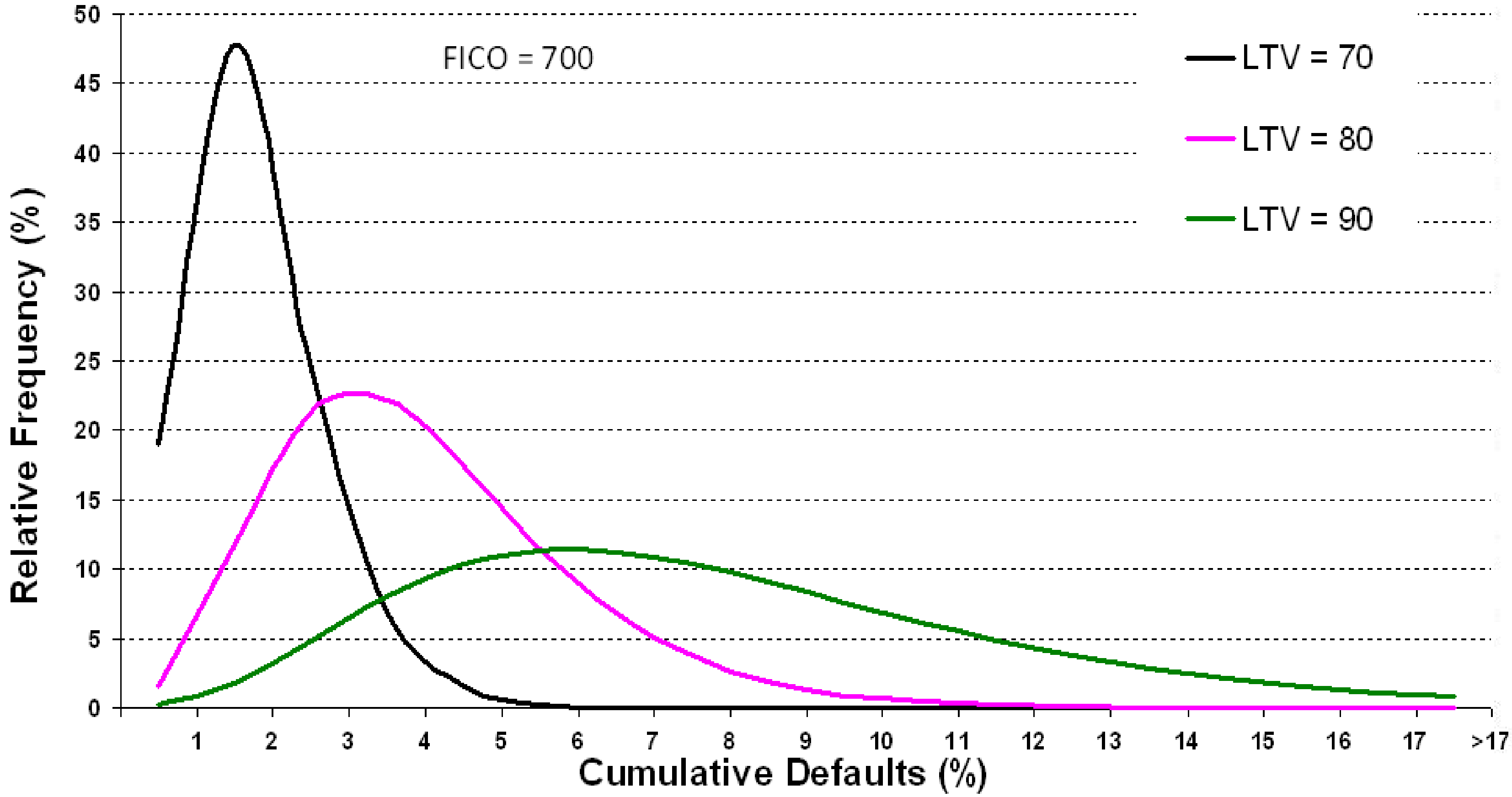
Envelope of Projected Cumulative Defaults



Source: Citi

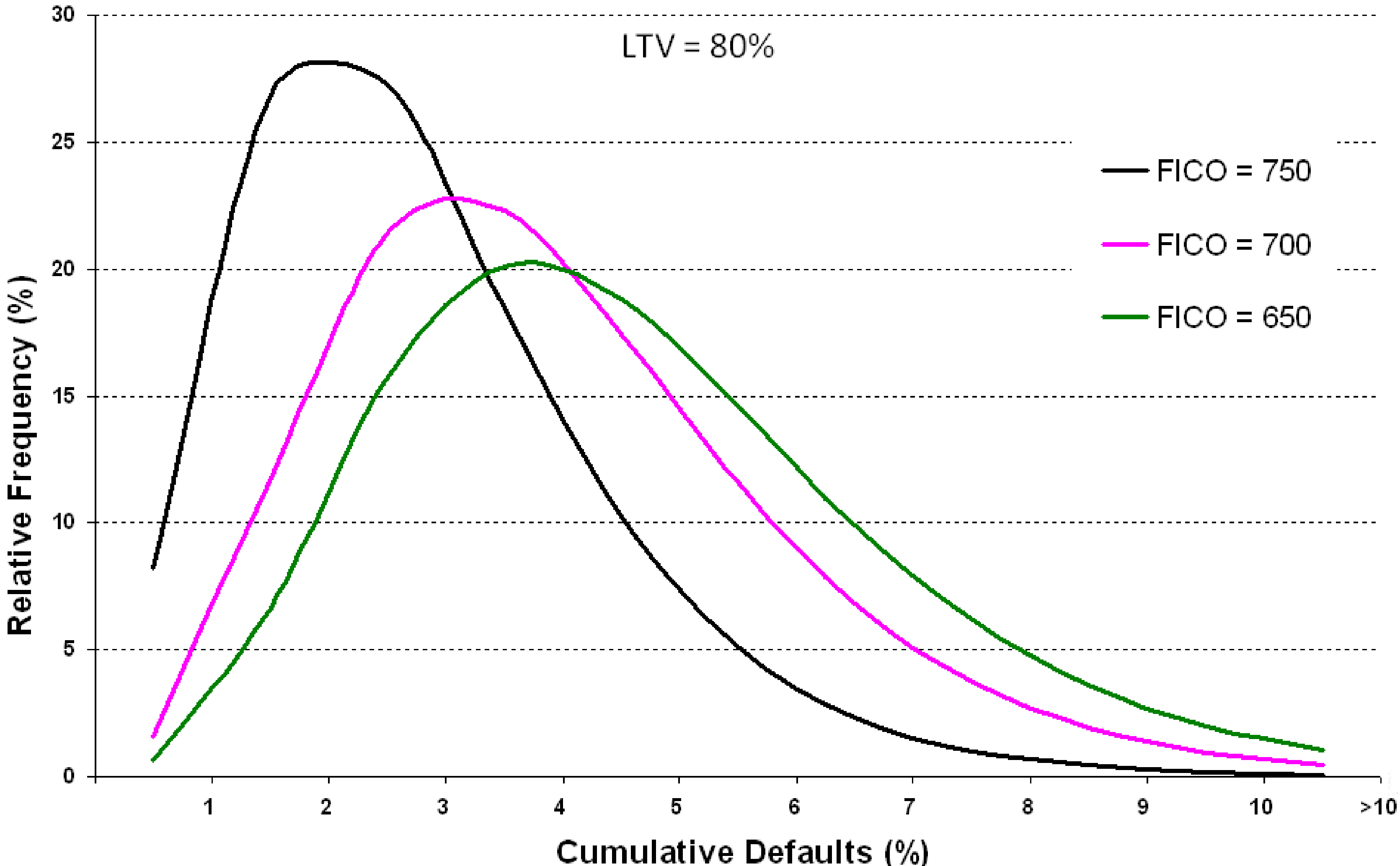
- Historical data on mortgage defaults can be used augment recent experience to model defaults in a post-bubble housing market;
- A summary of such data is give in *Evaluation of Credit-Sensitive Mortgage Securities,* Lakhbir Hayre, et al. Prudential Securities (1993)

Distribution of Projected Cumulative Defaults for New Loans with Varying Initial LTVs



Source: Citi

Distribution of Projected Cumulative Defaults for New Loans with Varying Credit Scores



Source: Citi

Analysis of STACR 2015-DN1

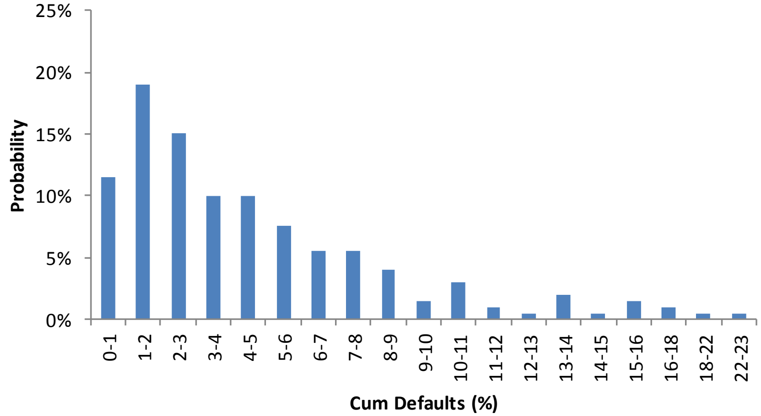
We look at STACR 2015-DN1 deal that includes a first loss piece and analyze it using our stochastic home price model for estimating credit-adjusted OAS.

Tranche	Curr Balance	Factor	Current CE	Coupon	Cpn Formula	Px (IDSI)	DM	Yield	WAL	PWD	CA-OAS	Cum Def
AH	24,667,153,903	0.934	4.79									
M1	227,192,588	0.988	3.73	1.42	1moL + 1.25	100.092	104	1.22	0.43	0	101	1.56
M2	230,000,000	1.000	2.67	2.57	1moL + 2.40	101.552	115	1.336	1.23	0	127	1.56
M3	345,000,000	1.000	1.07	4.32	1moL + 4.15	103.167	326	3.464	3.78	0	151	1.56
B	75,000,000	1.000	0	11.67	1moL + 11.50	92.119*	1000	10.398	8.30	26.3	557	1.56

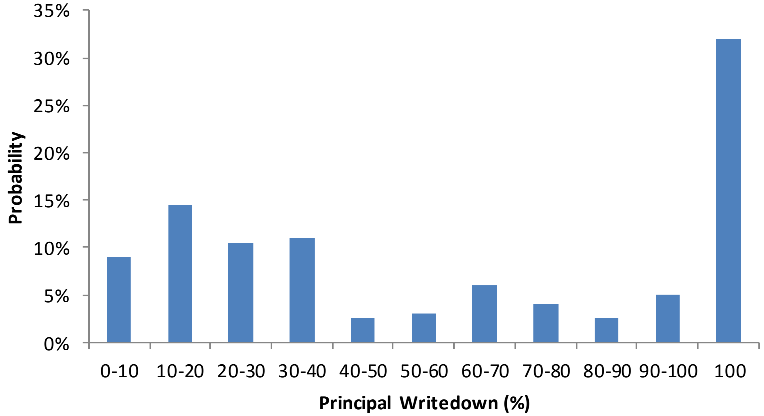
* Price based on DM=1000 bp in the base case scenario

For the base case scenario, cum defaults projected over a 10 year period is about 1.56%. For the credit-adjusted OAS analysis, 200 separate paths were run and the distribution of projected cum defaults is shown below.

Distribution of Cum Defaults



Distribution of PWD of Tranche B



Source: Citi

Credit-Adjusted OAS Analysis – Other Applications

HPA Duration

Given initial price PX and C-A OAS:

Shock Mean HPA Path +50bp: C-A OAS to Price ➔ PX(HPA+50bps)
Shock Mean HPA Path –50bp: C-A OAS to Price ➔ PX(HPA-50bps)

HPA Duration = 100* [PX(HPA+50bps)-PX(HPA-50bps]/PX

Separating Market and Credit Risk

R1= new interest rate level, H1= new HPA, PX = base price

Risk = PX – PX(R1,H1) = PX – PX(R1) + PX(R1) – PX(R1,H1)

= Change due to Rates + Additional change due to HPA

= Market Risk + Credit Risk

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